

## SAFETY DATA SHEET

Creation Date 30-April-2010

Revision Date 24-December-2021

Revision Number 4

### 1. Identification

**Product Name** tert-Butylchlorodiphenylsilane

**Cat No. :** AC187500000; AC187500100; AC187500500; AC187502500

**CAS-No** 58479-61-1  
**Synonyms** tert-Butyldiphenylsilyl chloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                  |              |                            |
|------------------------------------------------------------------|--------------|----------------------------|
| <b>Acute Inhalation Toxicity</b>                                 | Category 3   | (based on evolved HCl gas) |
| <b>Skin Corrosion/Irritation</b>                                 | Category 1 B |                            |
| <b>Serious Eye Damage/Eye Irritation</b>                         | Category 1   |                            |
| <b>Specific target organ toxicity (single exposure)</b>          | Category 3   |                            |
| Target Organs - Respiratory system.                              |              |                            |
| <b>Health Hazards Not Otherwise Classified</b>                   | Category 1   |                            |
| In contact with water, releases gases which are toxic if inhaled |              |                            |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Toxic if inhaled  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
In contact with water, releases gases which are toxic if inhaled



### Precautionary Statements

#### Prevention

Wash face, hands and any exposed skin thoroughly after handling  
Do not allow contact with water  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

#### Storage

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in a dry place

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                                  | CAS-No     | Weight % |
|--------------------------------------------|------------|----------|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | >95      |

## 4. First-aid measures

|                       |                                                                                                                                                                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>     | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Ingestion</b>      | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER                                                  |
| <b>Flash Point</b>                      | 149 °C / 300.2 °F                                                 |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | 480 °C / 896 °F                                                   |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

Corrosive material. Contact with water liberates toxic gas. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 1                   | 1                  | W                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. |
| <b>Environmental Precautions</b>            | Avoid release to the environment.                                                                                                                          |
| <b>Methods for Containment and Clean Up</b> | Do not expose spill to water. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe mist/vapors/spray. Do not allow contact with water. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Store under an inert atmosphere. Corrosives area. Incompatible Materials. Strong oxidizing agents.                                                               |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                            |
|----------------------------------|----------------------------|
| Physical State                   | Liquid                     |
| Appearance                       | Light yellow               |
| Odor                             | pungent                    |
| Odor Threshold                   | No information available   |
| pH                               | No information available   |
| Melting Point/Range              | No data available          |
| Boiling Point/Range              | 135 °C / 275 °F @ 0.7 mmHg |
| Flash Point                      | 149 °C / 300.2 °F          |
| Evaporation Rate                 | No information available   |
| Flammability (solid,gas)         | Not applicable             |
| Flammability or explosive limits |                            |
| Upper                            | No data available          |
| Lower                            | No data available          |
| Vapor Pressure                   | No information available   |
| Vapor Density                    | No information available   |
| Specific Gravity                 | 1.060                      |

|                                               |                                       |
|-----------------------------------------------|---------------------------------------|
| <b>Solubility</b>                             | No information available              |
| <b>Partition coefficient; n-octanol/water</b> | No data available                     |
| <b>Autoignition Temperature</b>               | 480 °C / 896 °F                       |
| <b>Decomposition Temperature</b>              | No information available              |
| <b>Viscosity</b>                              | 16 mPa.s at 25 °C                     |
| <b>Molecular Formula</b>                      | C <sub>16</sub> H <sub>19</sub> Cl Si |
| <b>Molecular Weight</b>                       | 274.86                                |

## 10. Stability and reactivity

|                                         |                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                            |
| <b>Stability</b>                        | Water reactive. Moisture sensitive.                                            |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Exposure to moist air or water.            |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                        |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                  |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

### **Component Information**

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                  | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------|------------|------------|------------|------------|------------|------------|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Insoluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

**UN-No** UN2987  
**Proper Shipping Name** Chlorosilanes, corrosive, n.o.s. consumer commodity  
**Hazard Class** 8  
**Packing Group** II

### TDG

**UN-No** UN2987  
**Proper Shipping Name** Chlorosilanes, corrosive, n.o.s.  
**Hazard Class** 8  
**Packing Group** II

### IATA

**UN-No** UN2987  
**Proper Shipping Name** Chlorosilanes, corrosive, n.o.s.  
**Hazard Class** 8  
**Packing Group** II

### IMDG/IMO

**UN-No** UN2987  
**Proper Shipping Name** Chlorosilanes, corrosive, n.o.s.  
**Hazard Class** 8  
**Packing Group** II

## 15. Regulatory information

### International Inventories

| Component                                  | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | -   | X    | X    | ACTIVE                                        | 261-282-0 | -      | -   |

| Component                                  | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------|------------|-------|------|------|------|------|------|-------|-------|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | -     | -    | -    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                  | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                  | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Silane, chloro(1,1-dimethylethyl)diphenyl- | 58479-61-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

#### Prepared By

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Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

#### Creation Date

30-April-2010

#### Revision Date

24-December-2021

#### Print Date

24-December-2021

#### Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**